

(Variant)

1.8. Let  $f(x_1, x_2)$  be a homothetic function. Show that its technical rate of substitution at  $(x_1, x_2)$  equals its technical rate of substitution at  $(tx_1, tx_2)$ .

Let  $f(x_1, x_2)$  be a homothetic function.

That is, we can define  $f(x_1, x_2) := h(g(x_1, x_2))$  where  $g: \mathbb{R}^2 \rightarrow \mathbb{R}^2$  is a HD1, and  $h: \mathbb{R}^2 \rightarrow \mathbb{R}^2$  is a monotone function.

$$TRS|_{(x_1, x_2)} = - \frac{\frac{\partial f}{\partial x_1}}{\frac{\partial f}{\partial x_2}} = - \frac{\frac{\partial h}{\partial g} \frac{\partial g}{\partial x_1}}{\frac{\partial h}{\partial g} \frac{\partial g}{\partial x_2}}$$

Take any  $t > 0$ .

As  $g$  is HD1,  $g(tx_1, tx_2) = t g(x_1, x_2) \Leftrightarrow \frac{\partial}{\partial x_i} g(tx_1, tx_2) = t \frac{\partial g}{\partial x_i} \quad (i=1,2)$ .

$$\begin{aligned} TRS|_{(tx_1, tx_2)} &= - \frac{\frac{\partial f}{\partial x_1}}{\frac{\partial f}{\partial x_2}} \bigg|_{(tx_1, tx_2)} = - \frac{\frac{\partial h}{\partial g} \frac{\partial}{\partial x_1} g(tx_1, tx_2)}{\frac{\partial h}{\partial g} \frac{\partial}{\partial x_2} g(tx_1, tx_2)} \\ &= - \frac{\frac{\partial h}{\partial g} (t \frac{\partial g}{\partial x_1})}{\frac{\partial h}{\partial g} (t \frac{\partial g}{\partial x_2})} \\ &= TRS|_{(x_1, x_2)}. \end{aligned}$$

2.3. Calculate explicitly the profit function for the technology  $y = x^a$ , for  $0 < a < 1$  and verify that it is homogeneous and convex in  $(p, w)$ .

For each  $(p, w) \in \mathbb{R}^2$ , the profit function is

$$\pi(p, w) := \sup_y \{ py - wx \mid y \leq x^a, 0 < a < 1 \}.$$

For given  $(p, w)$ , at  $(x^*, y^*)$  that achieves the maximized profit,

$$\begin{aligned} \frac{w}{p} &= a(x^*)^{a-1} \Leftrightarrow x^* = \left(\frac{ap}{w}\right)^{\frac{1}{1-a}} \\ &\Rightarrow y^* = \left(\frac{ap}{w}\right)^{\frac{a}{1-a}} \end{aligned}$$

$$\text{So } \pi(p, w) = p \left(\frac{ap}{w}\right)^{\frac{a}{1-a}} - w \left(\frac{ap}{w}\right)^{\frac{1}{1-a}}$$

i)  $\pi(p, w)$  is homogeneous in  $(p, w)$

Take any  $t > 0$ .

$$\begin{aligned} \pi(tp, tw) &= tp x^{*a} - tw x^* \\ &= t(px^* - wx^*) = t\pi(p, w). \end{aligned}$$

Thus,  $\pi$  is HD1.

ii)  $\pi(p, w)$  is convex in  $(p, w)$ .

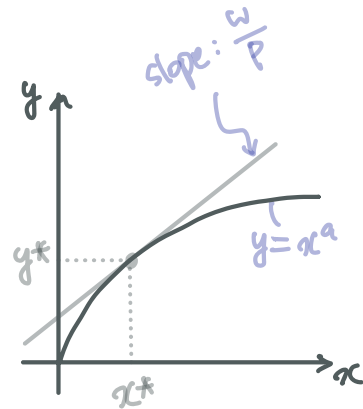
Take  $(p, w), (p', w') \in \mathbb{R}^2$ . Fix  $\lambda \in [0, 1]$ .

Define  $\bar{p} = \lambda p + (1-\lambda)p'$ ,  $\bar{w} = \lambda w + (1-\lambda)w'$ ,

$(\bar{x}^*, \bar{y}^*)$ : the optimal production point given  $(\bar{p}, \bar{w})$ .

$$\begin{aligned} &\lambda \pi(p, w) + (1-\lambda) \pi(p', w') \\ &= \lambda(px^* - wx^*) + (1-\lambda)(p'x'^* - w'x'^*) \\ &\geq \lambda(\bar{p}\bar{x}^* - \bar{w}\bar{x}^*) + (1-\lambda)(\bar{p}\bar{x}^* - \bar{w}\bar{x}^*) \\ &= (\lambda p + (1-\lambda)p')\bar{x}^* - (\lambda w + (1-\lambda)w')\bar{x}^* \\ &= \bar{p}\bar{x}^* - \bar{w}\bar{x}^* \\ &= \pi(\bar{p}, \bar{w}) \end{aligned}$$

Thus,  $\pi$  is convex in  $(p, w)$ .



2.4. Let  $f(x_1, x_2)$  be a production function with two factors and let  $w_1$  and  $w_2$  be their respective prices. Show that the elasticity of the factor share ( $w_2 x_2 / w_1 x_1$ ) with respect to  $(x_1 / x_2)$  is given by  $1/\sigma - 1$ .

$$\sigma = \frac{d \ln \frac{x_2}{x_1}}{d \ln |TRS|} \Rightarrow \frac{1}{\sigma} - 1 = \frac{d \ln |TRS| - d \ln \frac{x_2}{x_1}}{d \ln \frac{x_2}{x_1}}$$

At the optimum  $(x_1^*, x_2^*)$  that achieves the maximized profit,

$$\max_{(x_1, x_2)} p f(x_1, x_2) - (w_1 x_1 + w_2 x_2)$$

$$\Rightarrow \text{FOC: } \left. \begin{array}{l} p f_1 = w_1 \\ p f_2 = w_2 \end{array} \right\} \Leftrightarrow \underbrace{-\frac{f_1}{f_2}}_{\text{TRS}|_{(x_1^*, x_2^*)}} = -\frac{w_1}{w_2}$$

We can see that at the optimum,

$$\frac{1}{|TRS|} = \frac{w_2}{w_1}$$

The elasticity of factor share:

$$\begin{aligned} e &\equiv \frac{d \ln \left( \frac{w_2 x_2}{w_1 x_1} \right)}{d \ln \left( \frac{x_1}{x_2} \right)} = \frac{d \left( \ln \left( \frac{w_2}{w_1} \right) - \ln \left( \frac{x_1}{x_2} \right) \right)}{d \ln \left( \frac{x_1}{x_2} \right)} \\ &= \frac{d \left( \ln \left| \frac{1}{TRS} \right| - \ln \left( \frac{x_1}{x_2} \right) \right)}{d \ln \left( \frac{x_1}{x_2} \right)} \\ &= \frac{d \left( \ln |TRS| - \ln \left( \frac{x_2}{x_1} \right) \right)}{d \ln \left( \frac{x_2}{x_1} \right)} \\ &= \frac{1}{\sigma} - 1. \end{aligned}$$

2.7. The production function is  $f(x) = 20x - x^2$  and the price of output is normalized to 1. Let  $w$  be the price of the  $x$ -input. We must have  $x \geq 0$ .

- (a) What is the first-order condition for profit maximization if  $x > 0$ ?
- (b) For what values of  $w$  will the optimal  $x$  be zero?
- (c) For what values of  $w$  will the optimal  $x$  be 10?
- (d) What is the factor demand function?
- (e) What is the profit function?
- (f) What is the derivative of the profit function with respect to  $w$ ?

(a).  $\pi(w) = \max_{x \geq 0} f(x) - wx$

$$\Rightarrow \text{FOC: } f'(x) = w$$

$$\Leftrightarrow -2x + 20 = w \quad \text{with } x \geq 0.$$

If  $x > 0$ ,

for each  $x > 0$ , FOC gives  $w = -2x + 20$ .

(b) Optimal  $x$  is 0

$$\text{if } w = (-2)0 + 20 = 20.$$

(c) Optimal  $x$  is 10

$$\text{if } w = (-2)10 + 20 = 0.$$

(d) The factor demand is determined as

$$x^* = -\frac{1}{2}(w - 20) \quad \text{for the given } w \in \mathbb{R}$$

(e) The profit function:  
For each  $w \in \mathbb{R}$ ,

$$\begin{aligned}\pi(w) &= f(x^*) - wx^* = (20x^* - x^{*2}) - wx^* \\ &= -x^{*2} + (20-w)x^* \\ &= -\left(\frac{1}{4}(w-20)^2\right) + (20-w)\left(\frac{1}{2}(20-w)\right) \\ &= -\frac{1}{4}(20-w)^2 + \frac{1}{2}(20-w)^2 \\ &= \frac{1}{4}(20-w)^2\end{aligned}$$

(f) (Comparative Statics w.r.t  $w$ )

$$\begin{aligned}\frac{d\pi(w)}{dw} &= \frac{d}{dw} \left( \frac{1}{4}(20-w)^2 \right) \\ &= -\frac{1}{2}(20-w)\end{aligned}$$

┌ \*Note: By the Envelope Theorem,

$$\frac{d\pi(w)}{dw} = \frac{d}{dw} h(x;w) \Big|_{x=x^*}$$

(where  $h(x;w)$  is the objective fn.

$$= -x^*$$

i.e.,  $h(x;w) = f(x) - wx$  )

└

(Kreps)

9.6. Proposition 9.10 concerns a finite set of price-and-production-plan data that is meant to come from a single  $Z$ . Suppose we imagine a firm whose production possibilities expand through time. Specifically,  $Z_t$  for  $t = 1, 2, \dots$  gives the production possibilities for this firm at time  $t$ , and with  $Z_t \subseteq Z_{t+1}$  for all  $t$ . The data  $(p_1, z_1), (p_2, z_2), \dots, (p_T, z_T)$  are alleged to record the prices that face the firm and the production choice it made at times  $t = 1, 2, \dots, T$ , given this (expanding) production technology. Suppose we assume that the choice of the firm at time  $t$  maximizes the firm's profit at that time; that is,  $z_t \in \arg \max \{p_t \cdot z \mid z \in Z_t\}$ .<sup>1</sup> In the spirit of proposition 9.10, give necessary and sufficient conditions for the data  $(p_1, z_1), (p_2, z_2), \dots, (p_T, z_T)$  to be consistent with this story, for some sets  $Z_1, Z_2, \dots, Z_T$ .

**Proposition 9.10.** Given a finite list of prices and production plans,  $\{(p^n, z^n) \mid n = 1, \dots, N\}$  (where each  $p^n \in \mathbb{R}_{++}^k$  and  $z^n \in \mathbb{R}^k$ ), a set  $Z \subseteq \mathbb{R}^k$  exists such that  $z^n \in \arg \max \{p^n \cdot z \mid z \in Z\}$  for each  $n$  if and only if

$$p^n \cdot z^n \geq p^n \cdot z^m \text{ for each } n, m = 1, \dots, N. \quad (1)$$

Moreover, if (1) holds, the set  $Z$  works if and only if it satisfies

$$\{z^n \mid n = 1, \dots, N\} \subseteq Z \subseteq \{z \in \mathbb{R}^k \mid p^n \cdot z \leq p^n \cdot z^n, n = 1, \dots, N\}. \quad (2)$$

What additional condition do we need in order to have

$$\begin{cases} p^{t+1} \cdot z^{t+1} \geq p^{t+1} \cdot z^t & \dots \textcircled{1} \\ p^t \cdot z^t \geq p^t \cdot z^{t+1} & \dots \textcircled{2} \end{cases} \quad \text{for } \forall t=1, 2, \dots, T-1?$$

Since  $Z_t \subseteq Z_{t+1}$  for  $\forall t=1, \dots, T$ ,

We know that

$$\max \{p^t \cdot z \mid z \in Z_t\} \leq \max \{p^t \cdot z \mid z \in Z_{t+1}\} \dots \textcircled{a}$$

$p^t \cdot z^t \leq \max \{p^t \cdot z \mid z \in Z_{t+1}\}$  (assuming  $z^t \in \arg \max \{p^t \cdot z \mid z \in Z_t\}$ )

$$\max \{p^{t+1} \cdot z \mid z \in Z_t\} \leq \max \{p^{t+1} \cdot z \mid z \in Z_{t+1}\} \dots \textcircled{b}$$

$p^{t+1} \cdot z^t \leq p^{t+1} \cdot z^{t+1}$

From  $\textcircled{b}$ , the inequality  $\textcircled{1}$  is implied. However, for  $\textcircled{2}$ , we need something more than  $\textcircled{a}$ .

$p^t \cdot z^{t+1} \leq p^t \cdot z^t$  for sure if  $z^{t+1} \in Z_t$ .

If  $z^{t+1} \notin Z_t$  (that is,  $z^{t+1} \in Z_{t+1} \setminus Z_t$ ),  $p^t \cdot z^{t+1} \leq p^t \cdot z^t$

whenever  $p^t \cdot z^t = \max \{p^t \cdot z \mid z \in Z_t\} = \max \{p^t \cdot z \mid z \in Z_{t+1}\}$

<sup>1</sup>This is a very debatable assumption: Perhaps the firm's choice of production plan at time  $t$  affects the production possibilities it has at later times. Then its choice at time  $t$  should balance current profit against enhanced opportunities for profit in the future. But we do not take that possibility into account.

(i.e., given  $p^t$ ,  $z^t \in Z_t \subseteq Z_{t+1}$  is still a solution for maximum even in  $Z_{t+1}$ .)

This should hold for arbitrary  $t = 1, 2, \dots, T$ .

In short, what we need for the necessary & sufficient condition

to have the  $\{(p^t, z^t)\}_{t=1}^T$  be consistent with the story is that

$$p^t \cdot z^t = \max \{p^t \cdot z \mid z \in Z_t\} = \max \{p^t \cdot z \mid z \in Z_{t+1}\},$$

or equivalently,  $z^t \in \operatorname{argmax} \{p^t \cdot z \mid z \in Z_t\} \subseteq \operatorname{argmax} \{p^t \cdot z \mid z \in Z_{t+1}\}$